MARCOS JAVIER NIEVES ALVAREZ

Biotechnology · Bioinformatics · Bioprocess Engineering · Software

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PROFILE

Biotechnology undergraduate (Colegio Mayor de Antioquia) with hands-on experience in bioinformatics, bioprocess engineering and software development for the life sciences. I build reproducible analyses of biological data, edge-ML systems for plant disease detection, and my own web tools —including a production platform that teaches AI/ML to biotechnology students with Python running in the browser —. Daily working tools: Python, Linux, Git, Docker.

PROJECTS — SOFTWARE AND MACHINE LEARNING

Public repositories: **github.com/marcos-Nieves-24**

Snake Venom Analysis — reproducible biostatistical bioassay pipeline · Python, pandas, scikit-learn, Jupyter

- Reproducible analysis of cell-viability (MTT/XTT) data across 5 snake venoms × 3 cell lines × 5 concentrations × 2 time points: raw-data audit and reconstruction, exploratory analysis, inferential statistics and dose–response modelling.
- Clustering, machine-learning models and sensitivity analysis; 9 ordered executable notebooks plus reusable, typed and documented modules in `src/`.

InVitro-Code — web platform teaching AI/ML to biotechnology students · Next.js 16, TypeScript, MDX, Pyodide, Supabase, Clerk, FastAPI, Docker, Vercel

- **Production** platform with lessons, quizzes and in-browser executable Python labs (Pyodide v0.25); editorial content in MDX, FastAPI backend with multi-tenant isolation and row-level security, and gamification (streaks, achievements, leaderboard).
- 37 unit tests (Vitest) with >90% coverage, continuous deployment on Vercel, Spanish-language content.

Thyme × *Fusarium* spp. — factorial analysis of antifungal activity · Python, Jupyter, BDCA

- Statistical pipeline evaluating thyme (*Thymus* spp.) extracts against *Fusarium* spp. isolates: factorial ANOVA with interactions, dose–response curve adjustment, Tukey post-hoc testing, and automated figure and report generation. Design: extraction technique × isolate × concentration.

Other projects — Machine Learning Zoomcamp (DataTalksClub): regression and ensembles · RNA-seq-to-RNAi candidate pipeline (in development) · BioEncrypter, encryption and integrity tooling for sensitive biological data · virtual bioprocess simulator with PID control in C++.

RESEARCH EXPERIENCE

Research Student — Artificial Intelligence Research and Development Group (GIDIA), National University of Colombia, Faculty of Mines, Medellín · 2026 – Present

- Built the *Disease Detection Node*: an edge-ML system for plant disease detection, with CNN models (MobileNetV2, SqueezeNet) converted to TensorFlow Lite and inference on a Raspberry Pi 4 with USB camera and LED illumination.
- ML and REST API services orchestrated with Docker Compose, an MQTT broker and ESP32 firmware with LoRaWAN telemetry; clean architecture in the ML service (domain/application/adapters), type checking with mypy and ~290 automated tests.
- Annotation and structuring of ~3,000 drone images of fruit-tree crops (orange, avocado, mango) to generate segmentation polygons for the project's agricultural computer-vision dataset.

Research Student — Colegio Mayor de Antioquia · Mar 2026 – Present

- Bioplastic production from forage-sorghum stem residues (*Corpoica* JJT-18): physicochemical characterisation of the residue, extraction and analysis of biopolymers from lignocellulosic biomass, and design and standardisation of the production protocol.

Research Trainee — Colegio Mayor de Antioquia · Feb 2025 – Nov 2025

- Conventional and ultrasound-assisted enzymatic hydrolysis of purple yam starch (*Dioscorea alata*) obtained from anthocyanin-extraction residue; extraction yields and DNS quantification of reducing sugars.

International Research Intern — Universidad Autónoma de Tamaulipas, Mexico · Jun – Jul 2025

- Characterisation of a bioreactor for biosurfactant and bioplastic production: evaluation of operating parameters and bioprocess optimisation.

ADDITIONAL TRAINING

Transcriptomics: data processing and annotation — CCBIO, 2026
FASTQ, de novo assembly (Trinity), functional annotation (Trinotate), differential expression (DESeq2/edgeR), FastQC/MultiQC

Genomics and transcriptomics with Python and Linux — Lyn Biolearning, in progress
Workflow automation in Bash, libraries and genomic data analysis

Applied genomics and bioinformatics for the study of prokaryotes — Colegio Mayor de Antioquia, 2025
Bioinformatics tooling for the genomic study of prokaryotic organisms of scientific and industrial interest

PUBLICATIONS AND PRESENTATIONS

- **Poster** — Bioplastic production from forage-sorghum stem residues · **[conference and year to be added]**
- **Oral presenter** — "Characterisation of a bioreactor for biosurfactant production". 30th Summer of Scientific and Technological Research on the Pacific, Delfín Programme, Nuevo Nayarit, Mexico · Aug 2025
- **Exhibitor** — "Hydrogen production: molecular evolution and *in silico* modelling of hydrogenase enzymes of the genus *Pyrococcus*". 13th Biotechnology Fair, Colegio Mayor de Antioquia · Nov 2024

TECHNOLOGIES

Programming: Python (pandas, scikit-learn, NumPy, SciPy, FastAPI) · TypeScript · Bash · SQL · **Infrastructure:** Linux · Git · Docker · Vercel · **Edge ML:** TensorFlow Lite · MQTT · Raspberry Pi · ESP32/LoRaWAN · **Bioinformatics:** Trinity · Trinotate · DESeq2 · FastQC · MultiQC · **Web and data:** Next.js · MDX · Pyodide · Supabase (RLS) · Clerk · Tailwind CSS

LANGUAGES

Spanish — native · English — B2 (oral presentation before an international academic audience) · Portuguese — intermediate